

AI-Augmented Architectural Design: Enhancing Creativity and Workflow with Generative Models

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Figure 1: Generated design options using *image-to-image* (img2img) translations from student 3D models on left column.

Abstract

The application of Generative AI in fields involving subjective design evaluation, such as architectural design, is already proving effective in certain design phases. As technology advances, its applicability is expected to expand across broader phases of the design process. Beyond serving as an automation tool for production, AI demonstrates particular utility in the early design stages, where creativity is essential. Image-generation AI, trained on datasets labeled with subjective opinions from a critical mass of users, can capture and reflect broader aesthetic preferences, styles, and moods, aligning with a designer's intent. In a case study within an architectural design course, students incorporated AI not as a mere automation tool for manual labor but as a co-pilot—a means to break free from creative stagnation, explore different design personas within themselves, and push the boundaries of their architectural thinking.

CCS Concepts

• Applied computing → Architecture (buildings); Education.

Keywords

Architectural Design, Generative AI

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1 Overview

In this submission for the Educator's Forum, I introduce a design assignment that challenges students to develop architectural building designs by integrating visual outputs from generative AI techniques, appropriately selected for each phase of development. This project-based assignment encourages creative thinking and problem-solving for real-world implementation, blending speculative design with technology-driven exploration. Students incorporate AI-generated outcomes into the traditional architectural design process, beginning with concept ideation through abstract sketches, followed by the development of precise, geometrically defined 3D models that address both pragmatic and conceptual needs based on architectural programs and site requirements. The final phase involves refining their vision into compelling visual representations tailored for potential clients. The assignment can be completed individually or in groups, offering students an opportunity to explore AI's role as a co-pilot in design while maintaining a strong foundation in traditional architectural methodologies.

2 Materials

In the schematic design phase, *text-to-image* (text2img) AI tools have proven effective in helping students visually explore architectural forms that align with specific site conditions and building programs. Novice designers, who initially struggled to think beyond rigid rectangular structures, successfully used AI-generated imagery to conceptualize building envelopes that welcome pedestrian flow. Rather than replicating AI outputs, students used them as inspiration to escape their creative limitations and develop organic forms through drafting and constructing new geometries on themselves (Figure 2). During the design development phase, AI-assisted



Figure 2: *Text-to-image* (txt2img) AI outputs of green and curved architecture (top left) inspired the student to break creative limitations and develop a 3D CAD model with a more fluid, pedestrian-friendly form (top right) instead of his usual rectangular structures. *Image-to-video* (img2vid) techniques streamline the creation of walkthrough animations from static images, allowing students to explore dynamic visual storytelling and fostering new perspectives on spatial representation (bottom).

Table 1: Metadata in tabular format

Summary	Students develop original approaches to building design through visual communication with AI-generated outputs, using different techniques at various design phases.
Learning outcomes	Students practice creative problem-solving, explore non-traditional design generation methods, and develop innovative presentations.
Classification	Design, architecture, 2D/3D representations.
Audience	Upper-level undergraduate and graduate students. Goals can be adjusted based on skill level.
Dependencies	Varied based on the selected project.
Prerequisites	Foundational skills essential to the discipline.
Strengths	Scalable in scope, applicable to other design disciplines. Encourages non-traditional, AI-assisted sequential design development. Reduce production time, including rendering frames, to allow more time for creative enhancements.
Weaknesses	Require students to define their own goals and project requirements, unlike conventional design exercises where instructors set objectives.
Variants	Unlimited design possibilities and approaches.
Assessment	Evaluated using a rubric focused on quality and effectiveness. Open discussions with invited academics on presentations at reviews.

image-to-image (img2img) translation techniques helped bridge the gap between abstract concepts and concrete spatial configurations and visual presence in physical worlds. By applying Stable Diffusion (SD)’s ControlNet via the Automatic1111 GUI to perspective images based on CAD-generated 3D massing models, students refined materiality and visual identity related to spatial characteristics, adjusting proportions and enhancing the sense of place (Figure 1).

Unlike *text-to-image* generation, which often operates like a slot machine with unpredictable results, *image-to-image* translations allow for more structured collaboration between the designer and AI. At the presentation phase, *image-to-video* (img2vid) techniques streamline the creation of walkthrough animations from static images without the need for time-intensive rendering in software. Using tools like SD and Runway Gen-3, students explore dynamic visual storytelling by manipulating camera movements and object interactions. This process not only improves efficiency but also fosters new perspectives on spatial representation (Figure 2).

3 Conclusion

Based on the author’s experience before and after the adoption of AI in architectural education, AI has undeniably expanded students’ expressive capabilities. While the balance between AI augmentation and fundamental skill acquisition (e.g., CAD modeling, hand-drafting) warrants further study, post-course surveys indicate no significant loss of essential knowledge. AI’s role as a co-pilot holds significant potential across various disciplines, particularly in capstone projects where there is no single correct answer, and individual creative approaches must be developed within a limited timeframe. Through iterative engagement with AI, students can gain new insights and refine their design thinking. A pressing challenge for the near future is the AI’s generation of editable, manufacturable 3D models that can be seamlessly integrated into CAD workflows. While AI-based tools for generating 3D textured meshes and point clouds already exist, their direct applicability to real-world construction remains limited. However, ongoing research in this area suggests that solutions may emerge sooner than expected. When Zaha Hadid’s revolutionary competition entry in 1982 was introduced, it was a shocking revelation—at the time, few could conceive of architecture in such fluid, organic forms. Today, AI can generate countless variations of Zaha-inspired designs simply because such styles are embedded in its training data. Whether AI can truly generate unprecedented architectural forms beyond the accumulation of prior imagery remains an open question—one that challenges our understanding of AI’s role in originality and design innovation.